

CSCE 763: Digital Image Processing

Homework #1

Instructor: Spring 2026
University of South Carolina
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Due: 1:15 pm EST, Tuesday, Feb 3

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Problem 1: Smallest Discernible Dot (25 pts)

Thinking purely in geometric terms, estimate the diameter of the smallest printed dot that the eye can discern if the page on which the dot is printed is 0.5 m away from the eyes.

Assumptions:

- (a) The distance between the center of the lens and the retina along the visual axis is 14 mm.
- (b) The visual system ceases to detect the dot when the image of the dot on the fovea becomes smaller than the diameter of one receptor (cone) in that area of the retina.
- (c) The fovea can be modeled as a square array of dimensions $1.5 \text{ mm} \times 1.5 \text{ mm}$, and the cones (about 337,000 in total) and spaces between the cones are distributed uniformly throughout this array.

Problem 2: Adjacency of Image Subsets (25 pts)

Consider the two image subsets S_1 and S_2 shown below. For $V = \{1\}$, determine whether these two subsets are (a) 4-adjacent, (b) 8-adjacent, or (c) m-adjacent.

	S_1					S_2				
0	0	0	0	0	0	0	0	1	1	0
1	0	0	1	1	0	1	0	0	0	1
1	0	0	1	0	1	1	0	0	0	0
0	0	1	1	0	0	0	0	0	0	0
0	0	1	1	1	0	0	0	1	1	1

Problem 3: Shortest Paths (25 pts)

Consider the image segment shown below.

- (a) Let $V = \{0, 1\}$ and compute the lengths of the shortest 4-, 8-, and m-path between p and q . If a particular path does not exist between these two points, explain why.
- (b) Repeat for $V = \{1, 2\}$.

			(q)	
	3	1	2	1
	2	0	0	2
	1	2	1	1
(p)	1	0	1	2

Problem 4: Set Expressions (25 pts)

Give expressions for the sets shown shaded in the following figure in terms of sets A , B , and C .

In the figure: A is a rectangle (top), B is a triangle (bottom-left), C is a triangle (bottom-right). The three shapes overlap.

Analyzing Each Figure

Figure (i): The shaded region is the part of A that overlaps with B but not C , plus the part of A that overlaps with C but not B . This is the part of A that is in exactly one of B or C :

$$(A \cap B \cap C^c) \cup (A \cap C \cap B^c)$$

Equivalently: $A \cap (B \oplus C)$ where $\oplus$ denotes symmetric difference, or simply $A \cap (B \cup C) \cap (B \cap C)^c$.

Figure (ii): The shaded region includes the part of B not in A or C , plus the intersection of A and C (including overlaps with B). Looking at the shading pattern:

$$(A \cap C) \cup (B \cap A^c \cap C^c)$$

Figure (iii): The shaded region is the part of B and C that is not in A . That is, the union of B and C minus A :

$$(B \cup C) \cap A^c$$

Results

(i) $(A \cap B \cap C^c) \cup (A \cap C \cap B^c)$

(ii) $(A \cap C) \cup (B \cap A^c \cap C^c)$

(iii) $(B \cup C) \cap A^c$